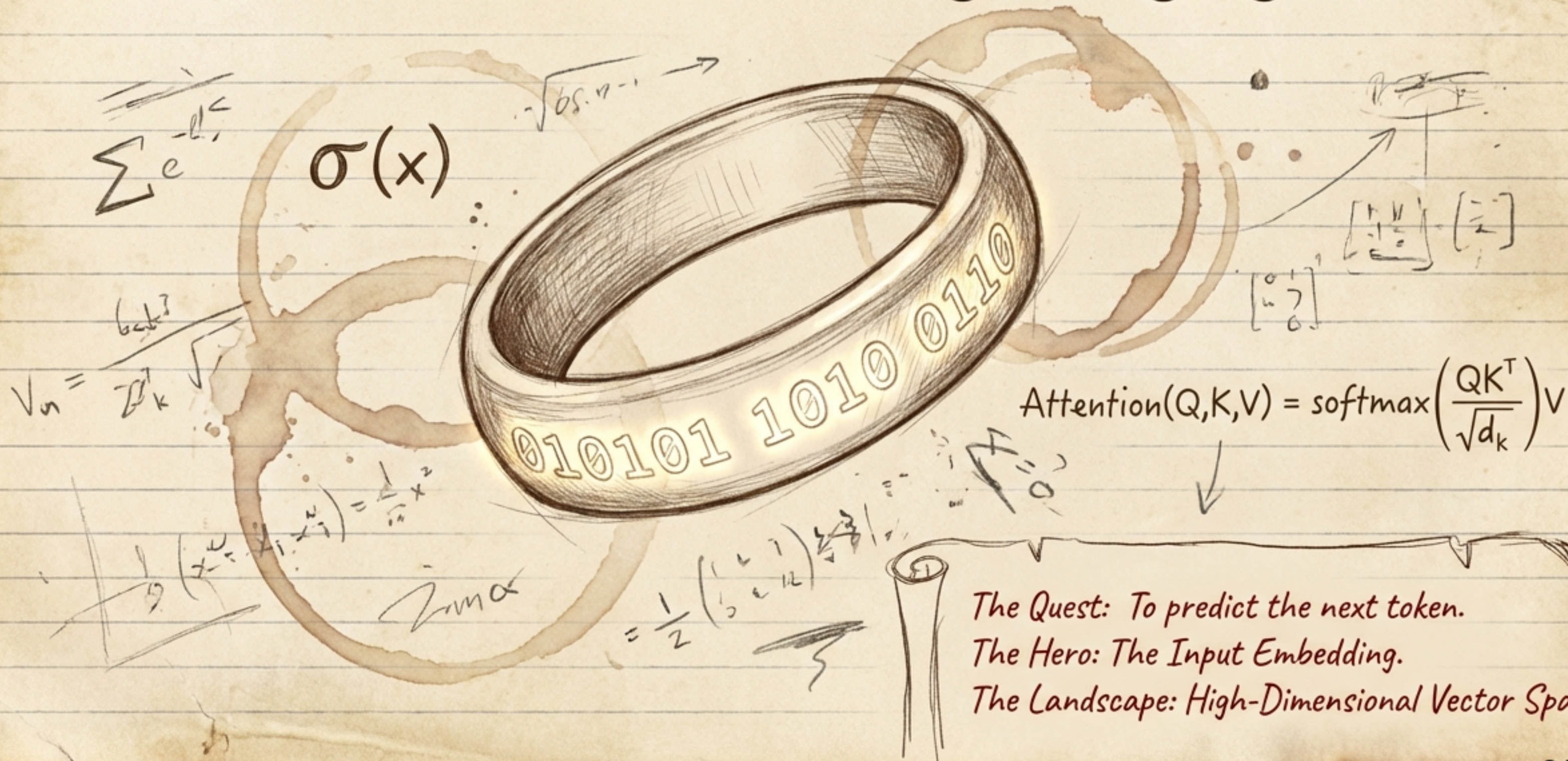


THE FELLOWSHIP OF THE TOKEN

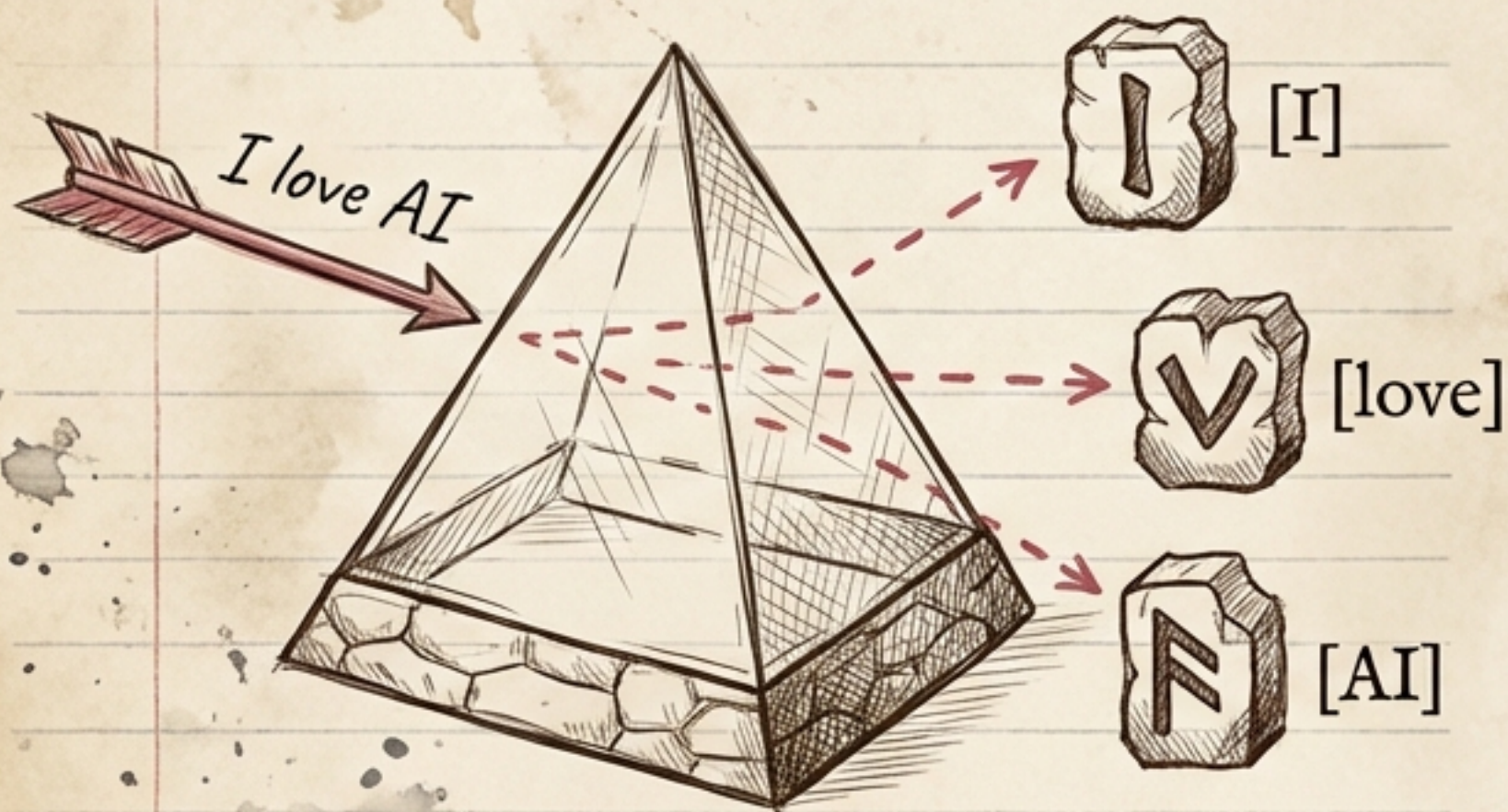
A Wizard's Field Guide to Large Language Models



THE WIZARD'S FIELD JOURNAL: TOKENIZATION & EMBEDDINGS

THE RUNES

IM Fell English SC

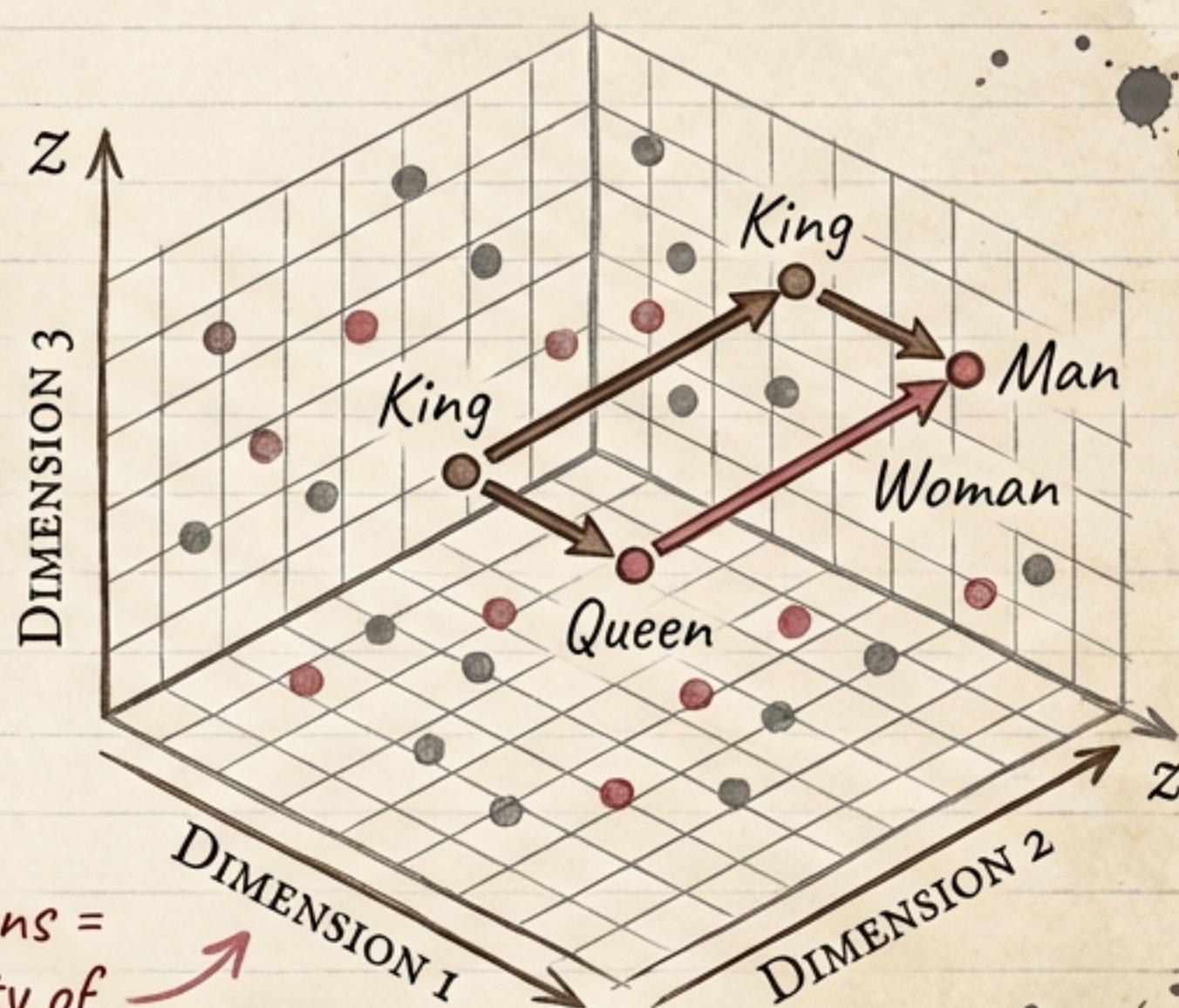


*The mundane tongue is broken,
forging tokens for the arcane.*

*768 Dimensions =
The complexity of
the character's soul.*

THE MAP

IM Fell English SC



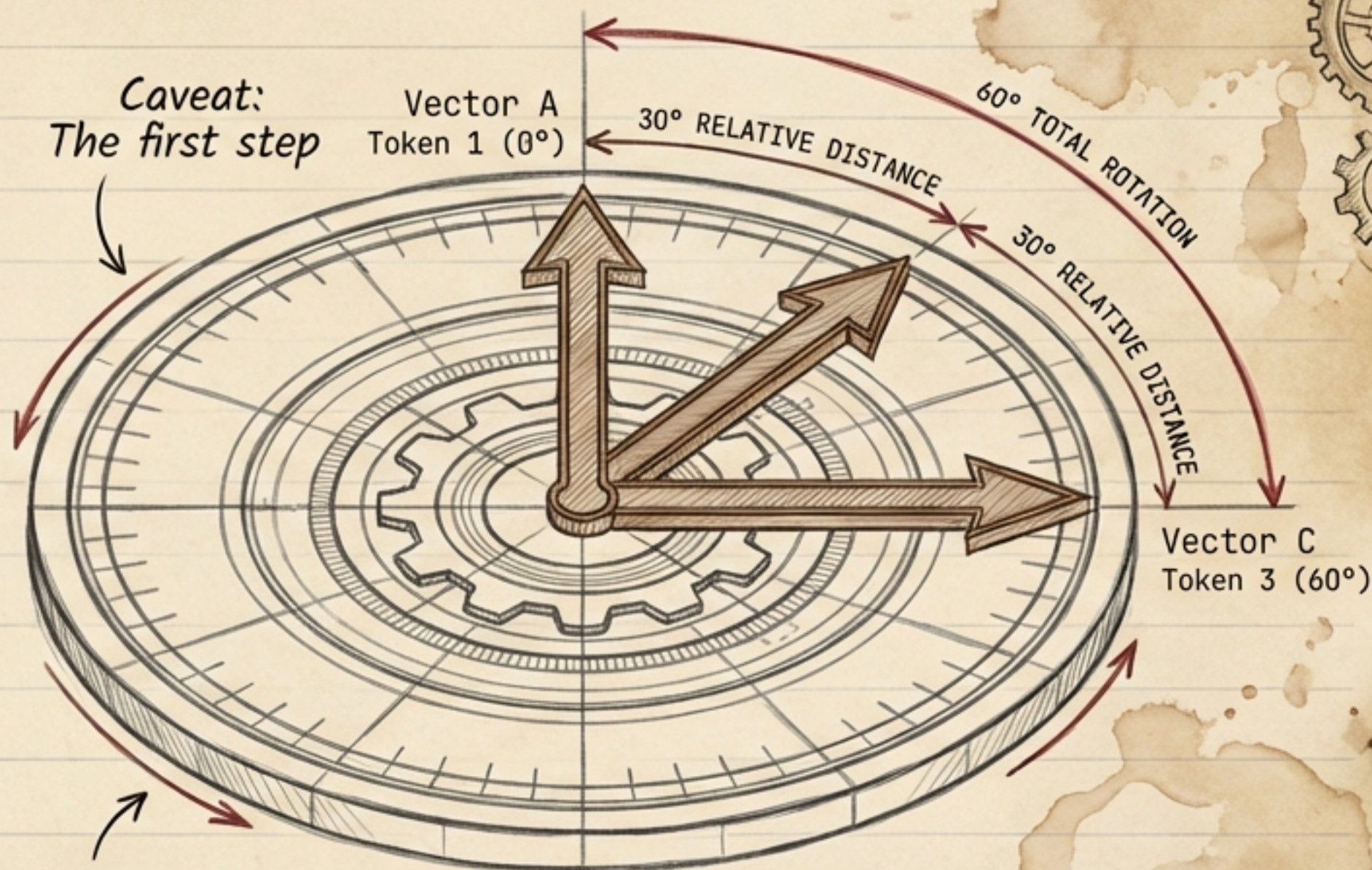
THE PATH TO MORDOR (POSITIONAL ENCODING)

THE PROBLEM:

Without position data, "The cat sat on the mat" is mathematically identical to "The mat sat on the cat".

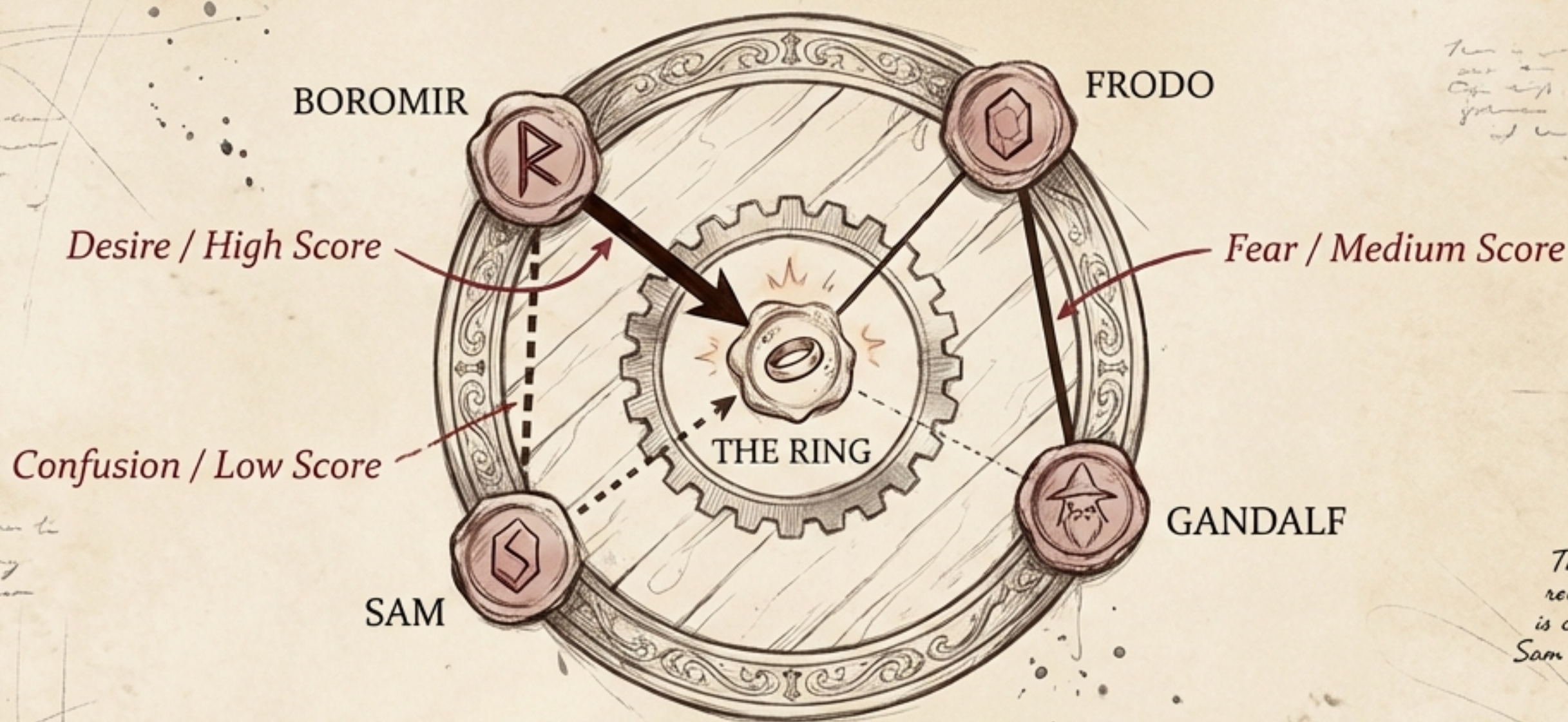
Rotation encodes relative distance. The order of the Fellowship matters.

*Caveat:
The first step*



*Caveat:
Rotation encodes relative distance directly into the vector. Token 3 is 30 degrees rotated from Token 1, representing adjacency regardless of absolute position in the journey.*

THE COUNCIL OF ELROND (SELF-ATTENTION)



MECHANISM: The model creates a weighted average of relationships.
Not every token is relevant to every other token.

The gaze of each character reveals their intent. Boromir is consumed by the Ring, while Sam is uncertain of the others.

THE SIGNAL FIRES (KEYS, QUERIES, VALUES)

QUERY (Q)

Signal
sent

KEY (K)

Response
initiated

VALUE (V)



What am I looking for?



Do I match the signal?

Signal
sent



What content do I transfer?

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

The animal didn't cross the street because it was too tired

45% 15% 45%

THE DIVERSE FELLOWSHIP (MULTI-HEAD ATTENTION)

Legolas / Long Range

Gimli / Structure

Gandalf / Meaning

The wizard arrived precisely when he meant to.

*Legolas / Long Range
Special Elite*

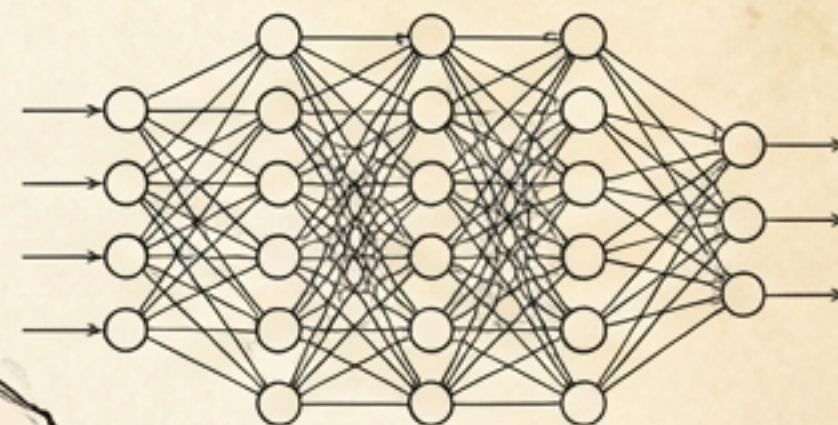
*Gimli / Structure
Special Elite*

*Gandalf / Meaning
Special Elite*

Concatenation = The Fellowship acting as one unit.

THE MINES OF MORIA (FEED-FORWARD NETWORKS)

The Knowledge Base



*The Thinking Phase:
Where facts are
stored in the weights.*



Input Layer (4096)

Hidden Layer (16384) - SwiGLU Activation

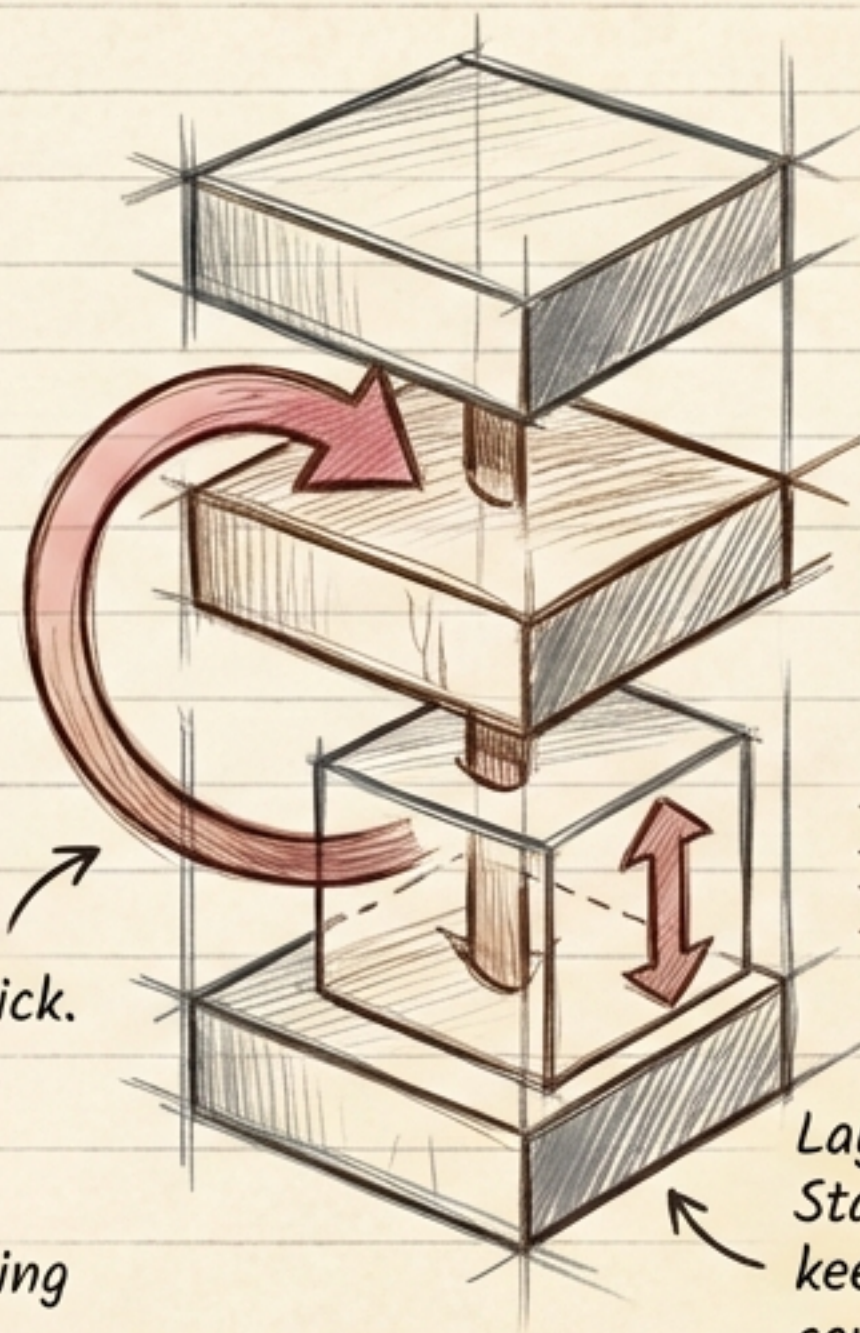
Output Layer

Galadriel's Light (Residuals & Norm)



Residual
Connection:
 $x + \text{Layer}(x)$

*Residual Connection:
The "Don't Forget" trick.
It allows gradients
(memories) to flow
through 100+
layers without vanishing
in the dark.*

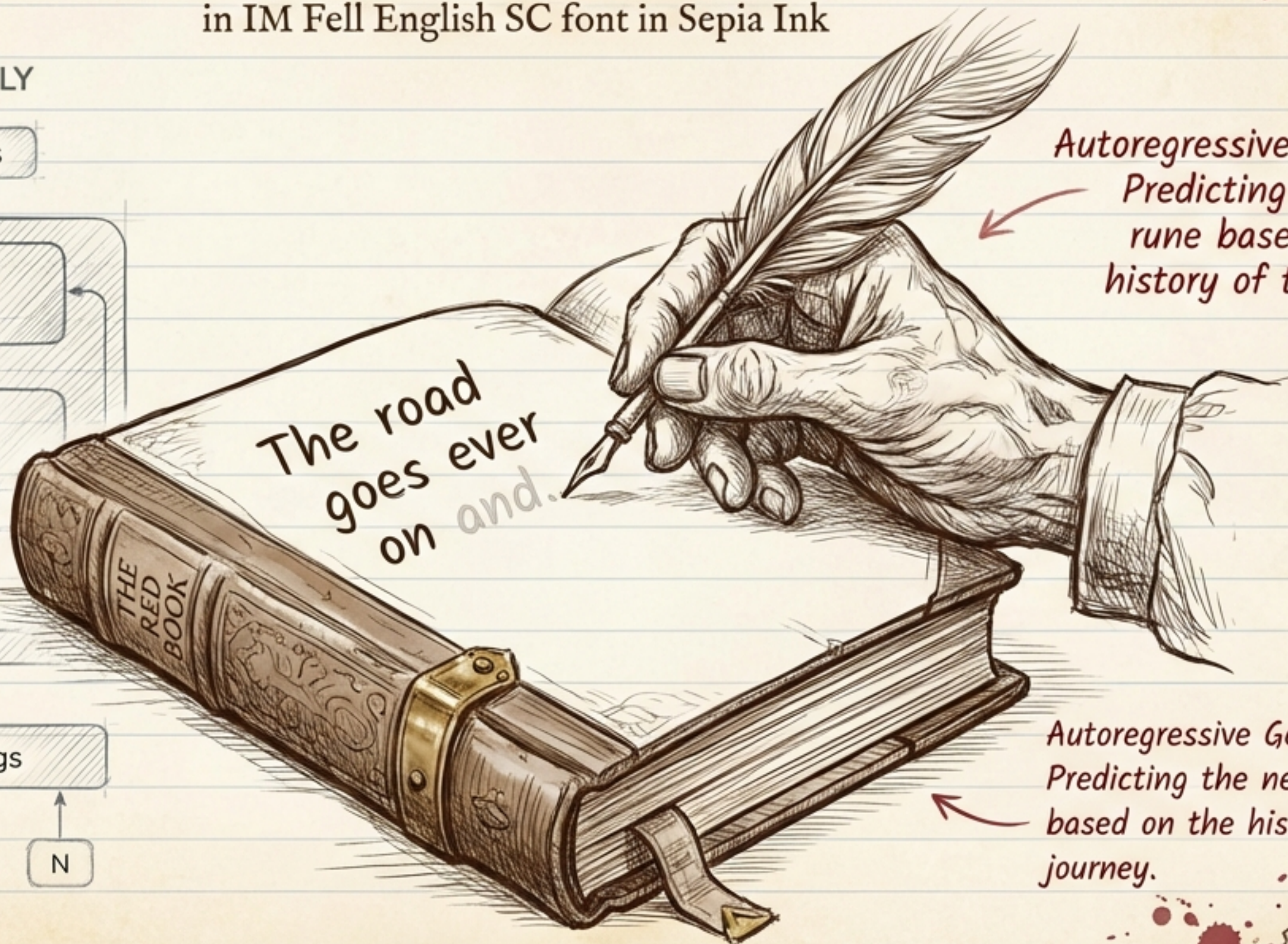
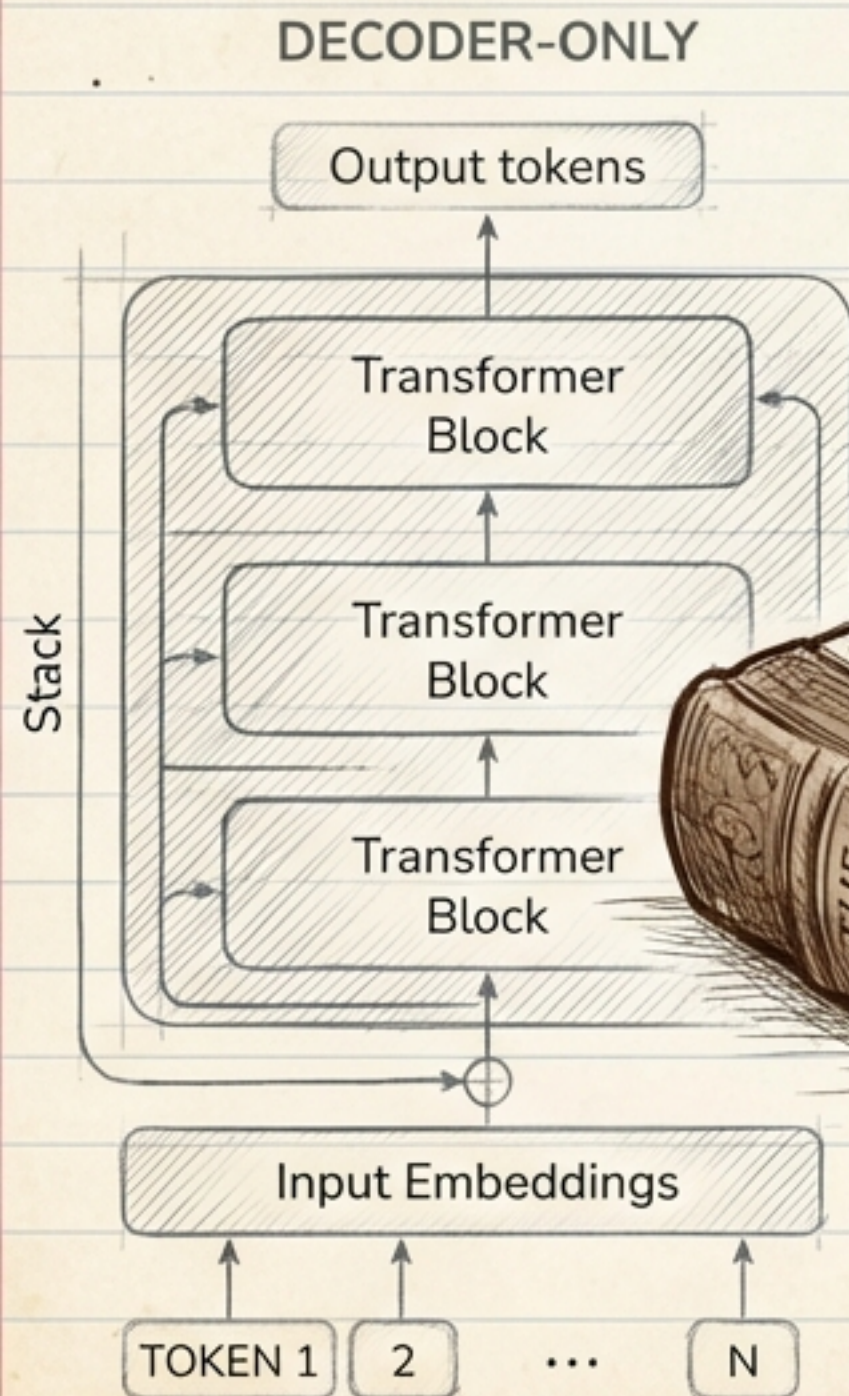


Layer Norm:
Bounded Variance

*Layer Norm:
Stabilizes the variance,
keeping the Fellowship
sane.*

THE SCRIBE (THE DECODER)

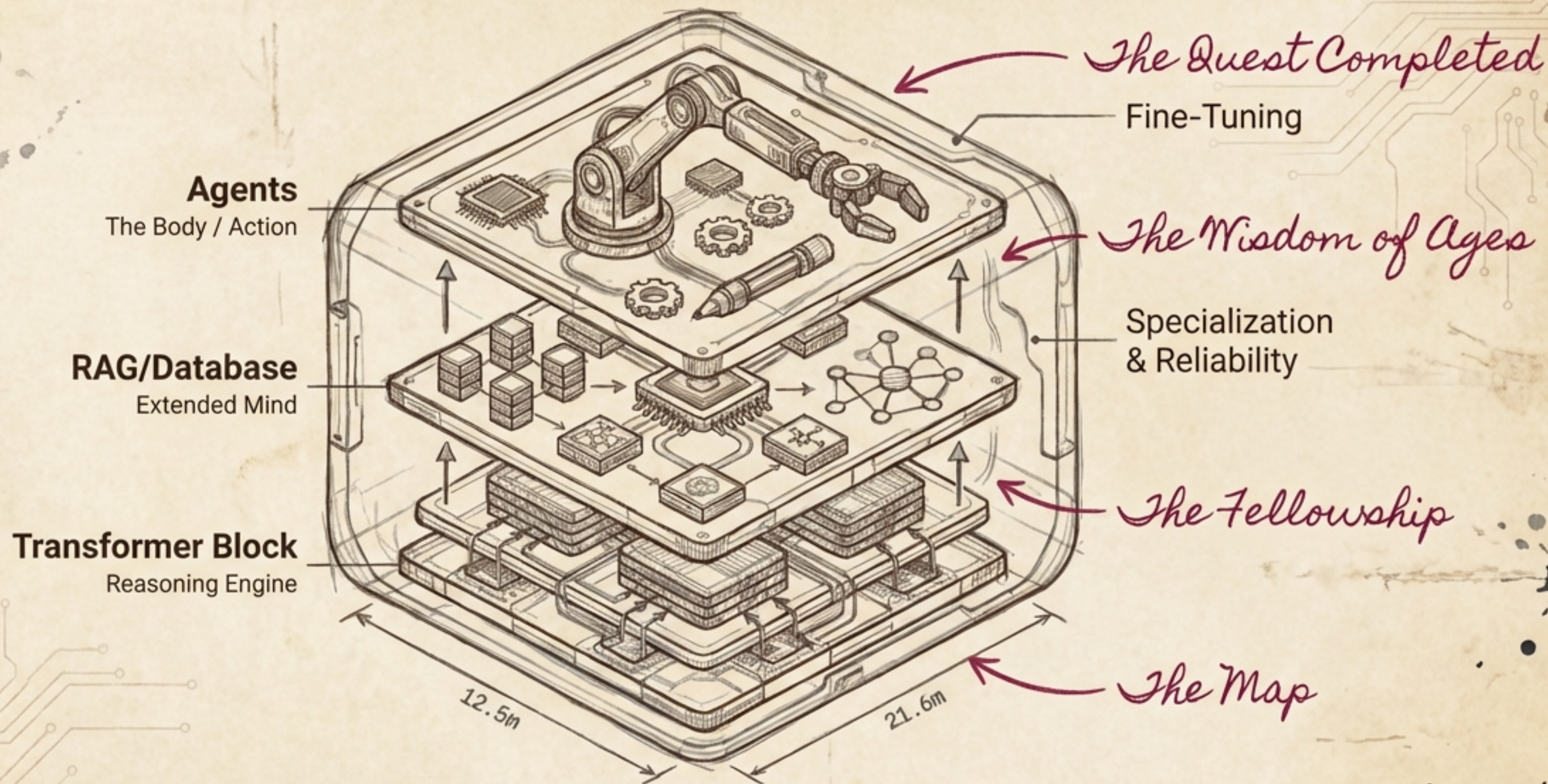
in IM Fell English SC font in Sepia Ink



*Autoregressive Generation:
Predicting the next
rune based on the
history of the journey.*

*Autoregressive Generation:
Predicting the next rune
based on the history of the
journey.*

THE RETURN OF THE KING (SUMMARY)



“The magic of AI is not magic. It is a Fellowship of math.”